Augmented matrix and Invertibility

Our goal is to find the inverse of a matrix using elementary row operations. This method has an advantage in case of matrices of large orders, when finding the determinant and minors become computationally inefficient. We discuss the method with an example.

Example 1. Find the inverse of the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 1 & 0 & 2 \end{pmatrix}.$$

We first form the augmented matrix with the given matrix on the left and the identity matrix on the right

$$(A|I) = \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 3 & 5 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{array}\right)$$

Using elementary row operations, our goal is to convert the left side matrix to identity matrix. The matrix that we obtain on the right is the required inverse. We proceed with the following row operations.

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 3 & 5 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3:R_3-R_1]{R_2:R_2-2R_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & -2 & -1 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2:(-1)R_2} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & -2 & -1 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_3:R_3+2R_2]{R_1:R_1-2R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & -3 & 2 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 3 & -2 & 1 \end{array}\right)$$

$$\xrightarrow[R_2:R_2-R_3]{R_1:R_1-R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -6 & 4 & -1 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 3 & -2 & 1 \end{array}\right)$$

Hence the required inverse is the matrix

$$A^{-1} = \begin{pmatrix} -6 & 4 & -1 \\ -1 & 1 & -1 \\ 3 & -2 & 1 \end{pmatrix}$$

What happens if we try to find the inverse of a non-invertible matrix by the above method? No matter what row operations we use, we will never be able to convert the left hand matrix into the identity matrix. For instance, try finding the inverse of the matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 2 \\ 3 & 3 & 2 \end{pmatrix}.$$

This gives us a way to detect whether a matrix is invertible or not by the augmented matrix method.

Exercise

Find the inverse, if it exists, of the following matrices

$$A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 & 2 \\ 3 & -3 & 1 \\ 3 & -3 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 1 & -1 \\ 3 & 1 & 1 \\ 1 & 2 & -1 \end{pmatrix},$$